

DTSA 5704

Managing, Describing, and Analyzing Data

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	Machine Learning Engineer (43.9 / 100)	31.4 / 100
SPECIALIZATION Statistical Quality Improvement		

31

CURRENT AVG SCORE
across 6 job roles

69

POTENTIAL SCORE
with recommended additions

+48

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

This course teaches the fundamentals of understanding data and the importance of correctly classifying data. This course covers how to describe data using descriptive statistics and R software, the four probability distributions commonly used in data analysis, steps for analyzing data sets using the appropriate probability distribution, and the basics of sampling error, sampling distributions, and errors in decision-making.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Machine Learning Engineer	43.9	69.7	+25.7	MEDIUM
Data Scientist	31.1	77.9	+46.7	HIGH
ML Engineer	30.9	58.3	+27.4	MEDIUM
Applied Scientist	30.2	77.8	+47.6	HIGH
AI Engineer	29.4	67.2	+37.8	HIGH
MLOps Engineer	22.8	61.6	+38.8	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 3 of 6 roles

systems

Deep Learning 1 of 6 roles

training

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- 1

HIGH

Add System Design coverage — required vocabulary in 3 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.
- 2

LOW

Optional: Deep Learning — would broaden appeal to one additional role. Candidate terms: training.

GAP KEYWORDS BY ROLE

Machine Learning Engineer 43.9 → 69.7 (+25.7 pts)

learning tools exploratory science models code applications machine

Data Scientist 31.1 → 77.9 (+46.7 pts)

toolkit scripting simply different social problems

ML Engineer 30.9 → 58.3 (+27.4 pts)

learning machine systems models problems technical building history

Applied Scientist 30.2 → 77.8 (+47.6 pts)

computational research reasoning literature problems models technical learning

AI Engineer 29.4 → 67.2 (+37.8 pts)

models learning technical code systems problems detection machine

MLOps Engineer 22.8 → 61.6 (+38.8 pts)

detection systems learning machine training tools models

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.